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RESEARCH ARTICLE

Microstimulation of V1 delays visually guided saccades: a parametric evaluation of delay fields

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Abstract Electrical microstimulation of macaque striate cortex (area V1) delays the execution of saccadic eye movements made to a visual target placed in the receptive field of the stimulated neurons. The region of visual space within which saccades are delayed is called a delay field. We examined the effects of changing the parameters of stimulation and target size on the size of a delay field. Rhesus monkeys were required to generate a saccadic eye movement to a punctate and white visual target presented within or outside the receptive field of the neurons under study. On 50% of trials, a train of stimulation consisting of 0.2-ms anode-first pulses was delivered to the neurons before the onset of the visual target. Stimulations were performed in the operculum at 0.9–2.0 mm below the cortical surface. It was found that increases in current (50–100 μA), pulse frequency (100–300 Hz), or train duration (75–300 ms) increased the size of a delay field and increases in target size (0.1°–0.2° of visual angle) decreased the size of a delay field. Delay fields varied in size between 0.1 and 0.6° of visual angle. These results are related to the properties of phosphenes induced by electrical stimulation of V1 in humans and compared to the interference effects observed following transcranial magnetic stimulation of human V1.

Keywords Occipital cortex · Electrical stimulation · Phosphenes · Transcranial magnetic stimulation · Rhesus monkeys

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Introduction

Electrical microstimulation of the striate cortex (area V1) interferes with the execution of visually guided saccadic eye movements (Schiller and Tehovnik 2001; Tehovnik et al. 2002, 2004, 2005; Tehovnik and Slocum 2003, 2005; Slocum and Tehovnik 2004). Stimulation immediately before the presentation of a visual target delays saccades made to the target when the target is located in the receptive field of the stimulated neurons (Tehovnik et al. 2004, 2005; Tehovnik and Slocum 2005). The excitability of the neurons mediating the delay is similar to that of neurons that subserve phosphenes generated by electrical stimulation of human V1 (Brindley and Lewin 1968; Dobelle and Mladejovsky 1974; Rushton and Brindley 1978; Tehovnik et al. 2004). Also, the size of the visual field within which saccades are delayed increases systematically with increases in the visual-field eccentricity coded by the stimulated V1 neurons (Tehovnik et al. 2005). This region of visual field affected by the stimulation is called a delay field. Based on these findings we have previously suggested that the study of delay fields can be used to make inferences about what monkeys see when stimulation is delivered to V1 (Tehovnik et al. 2005). The following is a study of the electrical and visual parameters that affect the size of delay fields.

Materials and methods

Subjects

Three adult rhesus monkeys (*Macaca mulatta*) C, H, and M were used. Access to water was limited before

each day of experimental testing. After testing the monkeys were allowed to drink until sated before being returned to the vivarium. Monkeys were provided for in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals and with the Guidelines of the Massachusetts Institute of Technology Committee on Animal Care.

Surgery

Monkeys were anaesthetized intravenously with pentobarbital (20 mg/kg) and prepared for aseptic surgery. A scleral search coil was implanted (Judge et al. 1980), and a stainless-steel post, used to restrain the head, was secured to the skull with titanium screws and acrylic cement. Subsequently, a recording chamber was fixed over the right V1.

Behavioural tasks

The animals, with head fixed in position, faced a computer monitor (Sony Multiscan E210) positioned 57 cm away in a dimly lit room. An animal was trained to first fixate a central spot (0.1° in diameter, 153 cd/m^2) and to then make a saccadic eye movement to a visual target (0.2° in diameter, 153 cd/m^2) that appeared at various locations on the monitor screen (with background luminance that could range from 7.7 to 76.7 cd/m^2). This task had two forms. In the first, the fixation spot remained on for 2–4 s and the animal had to maintain fixation until the spot was terminated and a target appeared. While the animal maintained fixation, a bar (153 cd/m^2) was swept across a restricted region of the monitor screen to map the receptive field of the neurons recorded from V1. After each sweep, the target appeared and the animal was rewarded with a drop of apple juice after a saccade had been made to the target. The orientation and size of the bar, as well as its direction of motion, were varied systematically until the location of the receptive field was determined (Fig. 1a, right).

For the second task, animals were required to fixate a spot of light (0.1° in diameter, 153 cd/m^2) for 600 ms, after which a visual target (0.1–0.2° in diameter, 153 cd/m^2) was presented within or outside the receptive-field location of the cells under study (Fig. 1c).

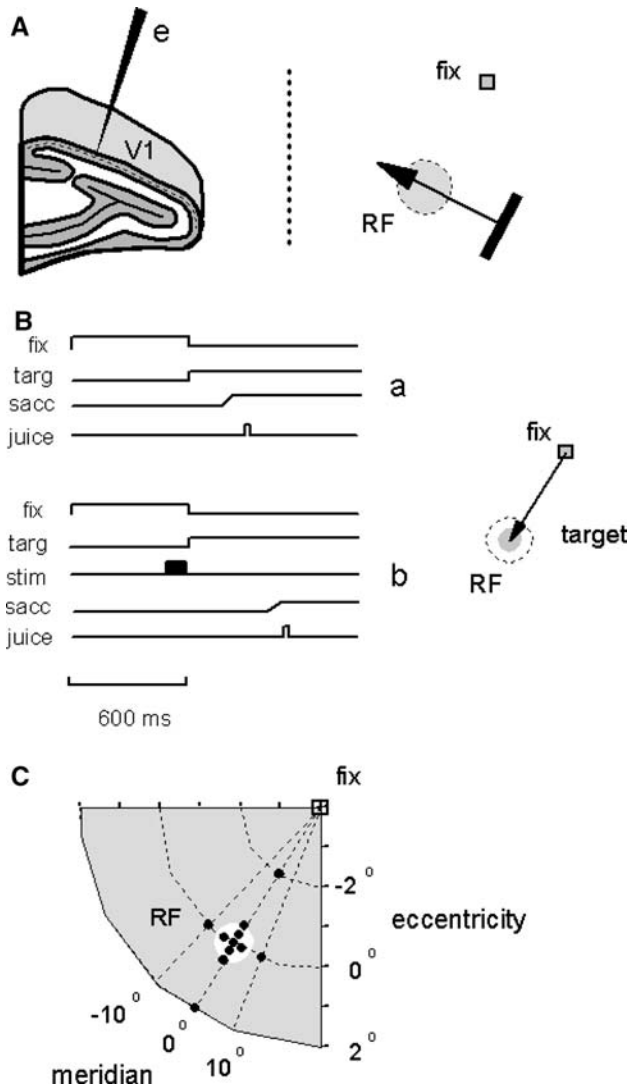

Fig. 1 Methods. (**a**, *left*) All electrodes (*e*) were lowered into V1 such that they penetrated V1 perpendicular to the cortical surface. (**a**, *right*) Receptive fields (*RF*) were mapped by moving a bar of light at different orientations across the visual field, as the monkey remained fixated (*fix*). **b** For all trials, an animal fixated a spot for 600 ms (fix), otherwise the trial was aborted and no juice reward delivered. Two types of trials were interleaved to study the stimulation-evoked delay effect. (a) Following fixation, a target (*targ*) was presented in (or outside) the receptive field of the stimulated neurons following the extinction of the fixation spot (also see illustration to right). The animal was required to generate a saccade (*sacc*) to the target within 500 ms to obtain a juice reward (*juice*). (b) Following fixation, a target (*targ*) was presented in (or outside) the receptive field of the stimulated neurons following the extinction of the fixation spot. Before the end of the fixation period, a train of electrical stimulation (*stim*) was delivered to the neurons. The animal was required to generate a saccade (*sacc*) to the target within 500 ms to obtain a juice reward (*juice*). **c** The target was presented at various locations with respect to the centre of the receptive field of the stimulated neurons. The target could occur from positive to negative 10° of meridian with respect the centre of the receptive field of the stimulated neurons and from positive to negative 2° of eccentricity with respect to the centre of the receptive field. The target used was brighter than background at 33% contrast (Michaelson) and was typically 0.2° in diameter. The fixation location (*fix*) of a monkey's centre of gaze and the receptive-field location (*RF*) of the stimulated neurons are indicated

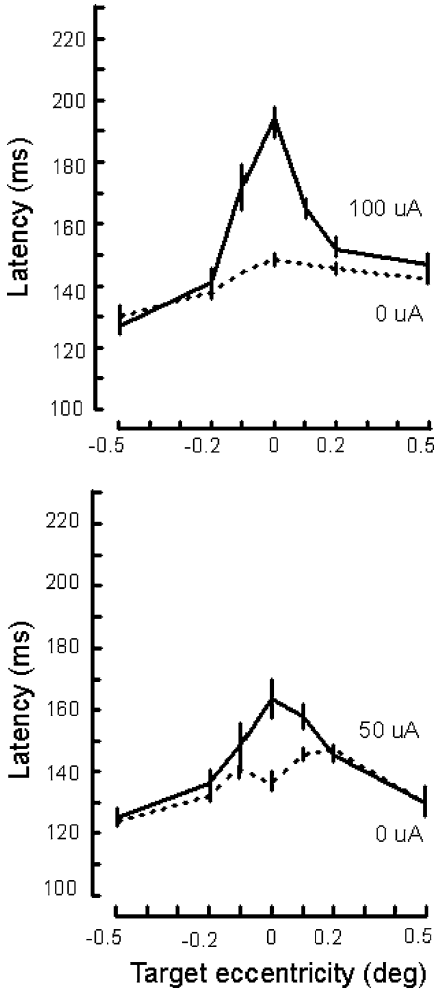

Fig. 2 Effect of stimulation on saccadic latency for different target positions at and outside of the receptive field of the stimulated neurons using two levels of current, 100 μA (*top*) and 50 μA (*bottom*). Within each panel (top or bottom), the latency of visually guided saccades to the target is plotted as a function of target eccentricity with respect to the receptive field of the stimulated neurons. A zero eccentricity along the *x*-axis indicates that the target and the receptive-field centre of the stimulated neurons were in register (see Fig. 1c). *Negative values* along the *x*-axis indicate target positions situated between the fixation position and receptive-field centre of the stimulated neurons. *Positive values* indicate target positions eccentric to the receptive-field centre of the stimulated neurons. The *solid curve* represents data from stimulation trials and the *dotted curve* represents data from non-stimulation trials. Each value is based on 20 trials. Standard errors are shown. The depth of stimulation was 1.5 mm below the cortical surface. The stimulation included anode-first pulses (at 0.2 ms duration) delivered at 200 Hz using a 100 ms train. The receptive field of the stimulated units was at 218° of meridian and at 2.6° of eccentricity. The target used was brighter than background at 33% contrast and was 0.2° in diameter. Data from monkey C. For other details see Materials and methods and Fig. 1b, c

The background luminance of the monitor was fixed at 76.7 cd/m^2, making the contrast of the visual target 33% (Michaelson). Monkeys received a juice reward for generating a saccade to the target within 500 ms following target onset (Fig. 1b(a)). During fixation, an animal was required to keep its eyes within a 0.5 × 0.5° window; otherwise the trial was aborted. During experimental sessions, a train of electrical stimulation was delivered to V1 on 50% of trials in a random order (Fig. 1b(b)). The train commenced before the end of the fixation period. The latency with which saccades were generated to the visual target constituted the primary data for the study.

Estimating electrode depth

For all electrode penetrations made into V1 (Fig. 1a, left), a standard method was used to deduce the electrode depth with respect to the top of cortex (Tehovnik et al. 2002, 2003). Once single units were encountered, their receptive field was mapped and the electrode was adjusted in depth until the unit activity was just detectable at the top of cortex. This point marked the initial depth at the top of V1. At the end of a test session, the depth at the top of V1 was re-measured by noting the point at which unit activity to visual stimuli became barely detectable as the electrode was slowly withdrawn. The estimated depth at the top of V1 was taken as the average of the initial and final depth estimates.

Data collection and analysis

A PDP-11/73 computer controlled the presentation of visual stimuli, the delivery of electrical stimulation, the collection of multiple-unit spikes, the acquisition and display of eye position (sampled at 200 Hz), and the delivery of juice.

Electrical stimulation and unit recording

Constant-current charge-balanced biphasic pulses were delivered to the brain via a monopolar glass-coated platinum–iridium electrode (at 0.3–0.5 MΩ tested at 1 kHz) using a Grass S88 stimulator attached to a pair of constant-current stimulus isolation units (Grass PSIU6B, Quincy, MA, USA). For each biphasic pulse, an anodal and cathodal pulse followed in immediate succession to produce anode-first pulses. Both pulses of a pair had the same amplitude and duration. Current was measured by the voltage drop across a 1,000 Ω resistor that was placed in series with the return lead of the stimulator. The current was monitored using an oscilloscope and was read as the amplitude of one pulse (anodal or cathodal) of a biphasic pair. The stimulation parameters used were as follows: current was 50 or 100 μA, pulse frequency varied from 100 to

300 Hz, and train duration varied from 75 to 300 ms. The pulse duration was fixed at 0.2 ms.

Electrodes were introduced perpendicular to the cortical surface with a hydraulic microdrive (Fig. 1a, left). To map visual receptive fields (Fig. 1a, right), action potentials were amplified (Bak A-1B, Germantown, MD, USA) and filtered (Krohn-Hite 3750, Horsham, PA, USA), isolated with a window discriminator (Bak DIS-1), and displayed on an oscilloscope (Tektronix TDS-210). The centre of a receptive field was defined as the target location yielding the greatest unit response to the same target as used in the stimulation experiments: i.e. the bright 0.2° target illuminated at 33% contrast (see Tehovnik et al. 2005).

Results

Stimulation sites

A total of 16 sites were studied in V1: 14 in monkey C, one in monkey D, and one in monkey H. The sites contained cells with receptive-field centres varying from 2.6 to 4.3° of eccentricity and from 215 to 259° of meridian. The depth of stimulation ranged from 0.9 to 2.0 mm below the cortical surface.

Effect of current, pulse frequency, and train duration on delay fields

Increases in current, pulse frequency, or train duration were found to increase the size of the visual field within which saccadic eye movements were delayed to a 0.2° visual target. The maximal delay occurred when the visual target was situated at the centre of the receptive field of the stimulate neurons.

Figure 2 shows the effect of increasing current from 50 to 100 μA for a 100-ms train of 200 Hz pulses delivered to a site in V1. The latency of saccades is plotted as a function of target eccentricity with respect to the centre of the visual receptive field of the stimulated neurons (as indicated by zero degrees of target eccentricity). The magnitude of the latency increase due to stimulation and the size of the field affected by the stimulation were greatest for the higher current level, i.e. 100 μA.

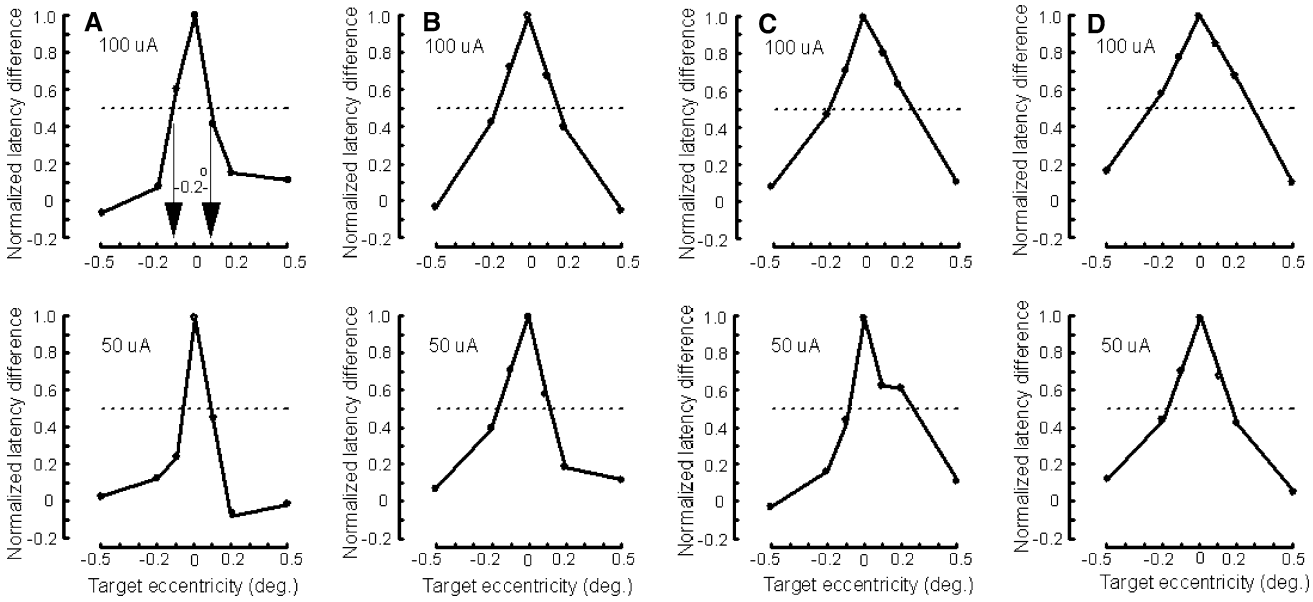

Fig. 3 Normalized latency difference is plotted as a function of target eccentricity for four stimulation sites (**a**–**d**) using two levels of current, 100 μA (*top*) and 50 μA (*bottom*). Data in **a** are derived from Fig. 2. Each curve in a panel was determined by subtracting the latency of saccades during non-stimulation trials from the latency of saccades during stimulation trials to yield a latency difference. The latency-difference data were then normalized such that the values spanned the range from zero to one. Normalized latency-difference values less than zero indicate that the saccadic latencies during stimulation were less than those on non-stimulation trials. The size of a delay field was measured by noting the negative and positive target-eccentricity values at 50% of maximal latency difference represented by the *dotted horizontal line*. The size of the delay field derived from the data in **a**, *top*, is 0.2° of visual angle (marked by the *arrows* in the figure; size = [(0.09 − (−0.11))] degree). The size of the delay field was always greater when using 100 μA than when using 50 μA: (**a** 0.2° vs. 0.16°; **b** 0.34° vs. 0.29°; **c** 0.44° vs. 0.34°; **d** 0.56° vs. 0.34°). For all sites, the stimulation was made up of anode-first pulses (at 0.2 ms duration) delivered at 200 Hz using a 100 ms train. The target used was brighter than background at 33% contrast and was 0.2° in diameter. **a** and **b** are from monkey C; **c** is from monkey H; **d** is from monkey M. For other details see Materials and methods and Fig. 1b, c

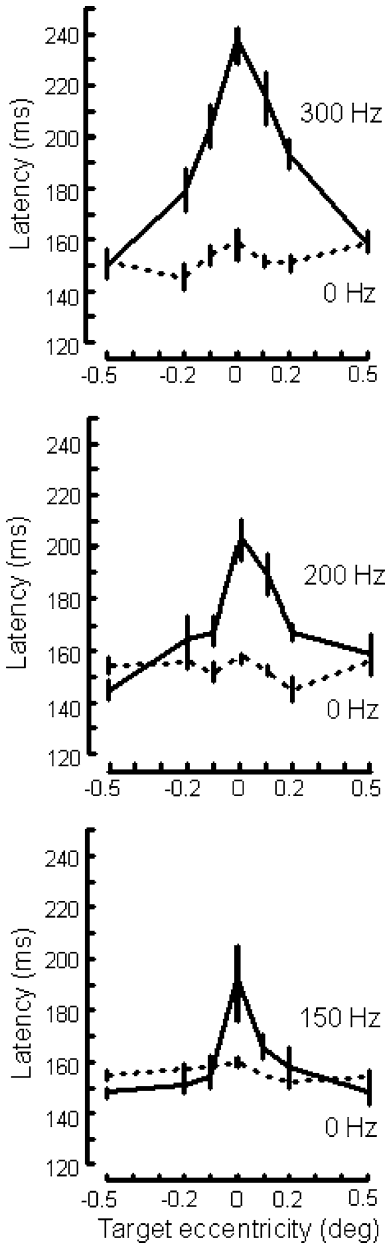

Fig. 4 Effect of stimulation on saccadic latency for different target positions at and outside of the receptive field of the stimulated neurons using three levels of pulse frequency, 300 Hz (*top*), 200 Hz (*middle*), and 150 Hz (*bottom*). Within each panel (*top*, *middle*, or *bottom*), the latency of visually guided saccades to the target is plotted as a function of target eccentricity with respect to the receptive field of the stimulated neurons. A zero eccentricity along the *x*-axis indicates target and the receptive-field centre of the stimulated neurons were in register (see Fig. 1c). *Negative values* along the *x*-axis indicate target positions situated between the fixation position and receptive-field centre of the stimulated neurons. *Positive values* indicate target positions eccentric to the receptive-field centre of the stimulated neurons. The *solid curve* represents data from stimulation trials and the *dotted curve* represents data from non-stimulation trials. Each value is based on 20 trials. Standard errors are shown. The depth of stimulation was 1.75 mm below the cortical surface. The stimulation included 100 μA anode-first pulses (at 0.2 ms duration) delivered using a 100 ms train. The receptive field of the stimulated units was at 238° of meridian and at 3.3° of eccentricity. The target used was brighter than background at 33% contrast and was 0.2° in diameter. Data from monkey C. For other details see Materials and methods and Fig. 1b, c

The size of a delay field was determined by noting the latency difference between stimulation and non-stimulation trials as a function of target eccentricity. These data were normalized such that the maximal latency difference was set to one and all other values became less than 1 (e.g. Fig. 3a, top). Zero values indicate that the latency of saccades on stimulation and non-stimulation trials was the same; positive values indicate that the latency of saccades on stimulation trials was greater than the latency of saccades on non-stimulation trials; and negative values indicate that the latency of saccades on stimulation trials was less than the latency of saccades on non-stimulation trials. The size of a delay field was measured by summing the negative and positive eccentricity values at which the normalized latency difference curves were 50% of peak. Such a measurement is illustrated in Fig. 3a, top, by the arrows. In this case, using a 100 μA current the size of the delay field was 0.2° of visual angle. For the four sites tested (Fig. 3a–d), the size of the delay field was greater when using 100 μA than when using 50 μA, respectively: a: 0.2° vs. 0.16°; b: 0.34° versus 0.29°; c: 0.44° vs. 0.34°; d: 0.56° vs. 0.34°). The data in Fig. 3a were derived from Fig. 2.

Figure 4 shows the effect of increasing the frequency of pulses from 150 to 300 Hz for a 100-ms train of 100-μA pulses delivered to a site in V1. The latency of saccades is plotted as a function of target eccentricity with respect to the centre of the visual receptive field of the stimulated neurons. The magnitude of the latency increase due to stimulation and the size of the field affected by the stimulation were greatest for the highest pulse frequency (300 Hz) and least for the lowest pulse frequency (150 Hz). For the four sites tested (Fig. 5a–d), the size of the delay field was greatest for the highest pulse frequency, intermediate for the midlevel pulse frequency, and least for the lowest pulse frequency, respectively: a: 0.38° versus 0.28° versus 0.12°; b: 0.27° versus 0.22° versus 0.15°; c: 0.34° versus 0.16° versus 0.12°; d: 0.28° versus 0.21° versus 0.12°. The data in Fig. 5a were derived from Fig. 4.

Figure 6 shows the effect of increasing the train duration from 75 to 150 ms for 100 μA pulses delivered at 200 Hz to a site in V1. The latency of saccades is plotted as a function of target eccentricity with respect

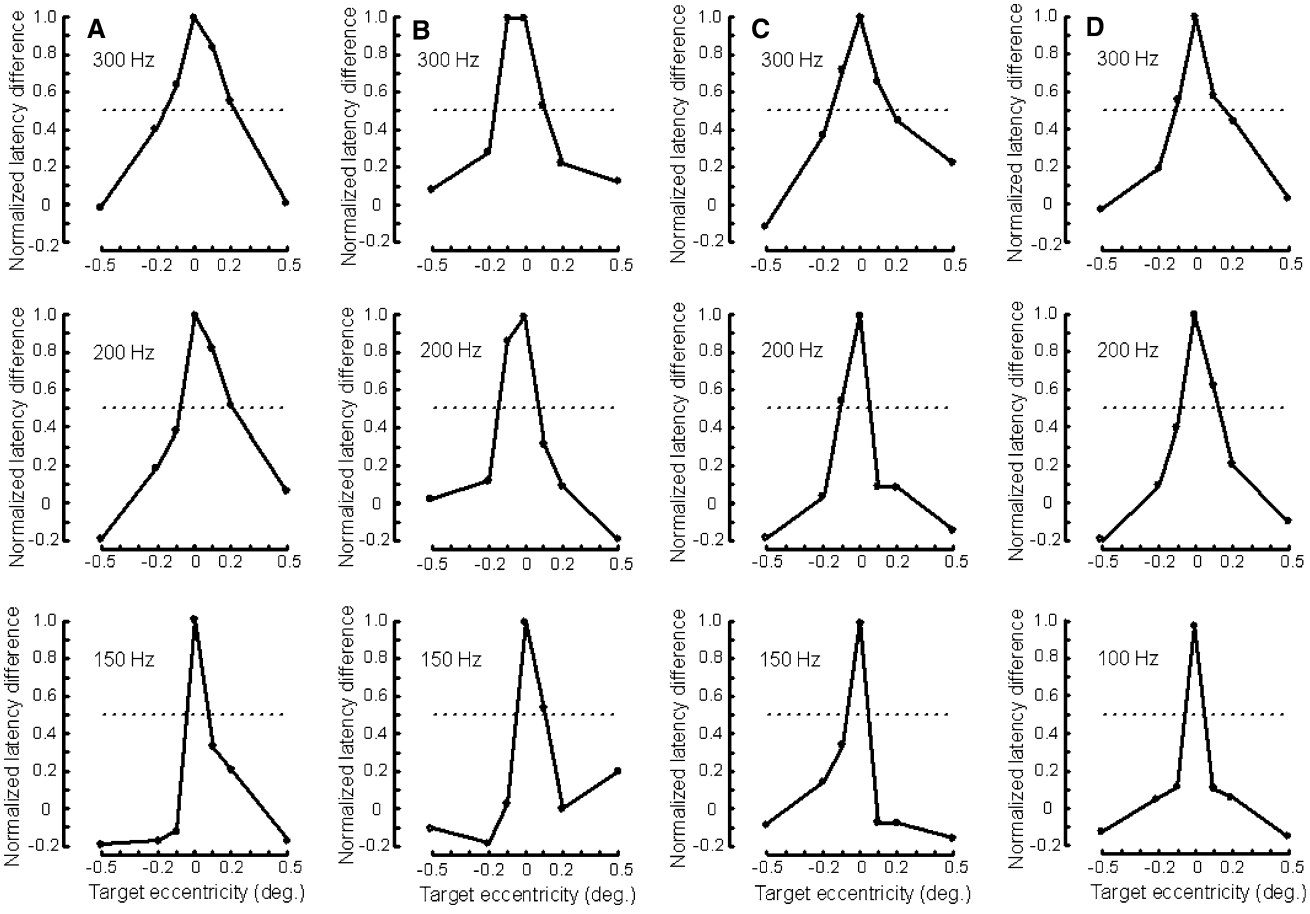

Fig. 5 Normalized latency difference is plotted as a function of target eccentricity for four stimulation sites (**a**–**d**) using three levels of pulse frequency, 300 Hz (*top*), 200 Hz (*middle*), and 150 or 100 Hz (*bottom*). Data in **a** are derived from Fig. 4. Each curve in a panel was determined by subtracting the latency of saccades during non-stimulation trials from the latency of saccades during stimulation trials to yield a latency difference. The latency-difference data were then normalized such that the values spanned the range from zero to one. Normalized latency-difference values less than zero indicate that the saccadic latencies during stimulation were less than those on non-stimulation trials. The size of a delay field was measured by noting the negative and positive target-eccentricity values at 50% of maximal latency difference represented by the *dotted horizontal line*. See Fig. 3 for other details. The size of the delay field was greatest for the highest pulse frequency, intermediate for the intermediate pulse frequency, and least for the lowest pulse frequency: (**a** 0.38° vs. 0.28° vs. 0.12°; **b** 0.27° vs. 0.22° vs. 0.15°; **c** 0.34° vs. 0.16° vs. 0.12°; **d** 0.28° vs. 0.21° vs. 0.12°). For all sites, the stimulation included 100 μA anode-first pulses (at 0.2 ms duration) using a 100 ms train. The target used was brighter than background at 33% contrast and was 0.2° in diameter. All data from monkey C. For other details see Materials and methods and Fig. 1b, c

to the centre of the visual receptive field of the stimulated neurons. The magnitude of the latency increase due to the stimulation did not vary systematically with increases in train duration, but the size of the field affected by the stimulation was greatest for the longest train duration (150 ms) and least for the shortest train duration (75 ms). For the four sites tested (Fig. 7a–d), the size of the delay field was greatest for the longest train duration, intermediate for the midlevel train duration, and least for the shortest train duration, respectively: a: 0.37° versus 0.19° versus 0.17°; b: 0.44° versus 0.29° versus 0.21°; c: 0.36° versus 0.32° versus 0.21°; d: 0.34° versus 0.27° versus 0.23°. The data in Fig. 7a were derived from Fig. 6.

Effect of target size on delay fields

Increasing the size of the visual target from 0.1 to 0.2° of visual angle was found to decrease the size of the delay field. In these experiments, the stimulation was made up of 100 μA pulses delivered at 200 Hz for a duration of 100 ms.

Figure 8 shows the effect of increasing target size. The latency of saccades is plotted as a function of target eccentricity with respect to the centre of the visual receptive field of the stimulated neurons. The magnitude of the latency increase due to the stimulation did not vary systematically with increases in target size, but the size of the field affected by the stimulation was least

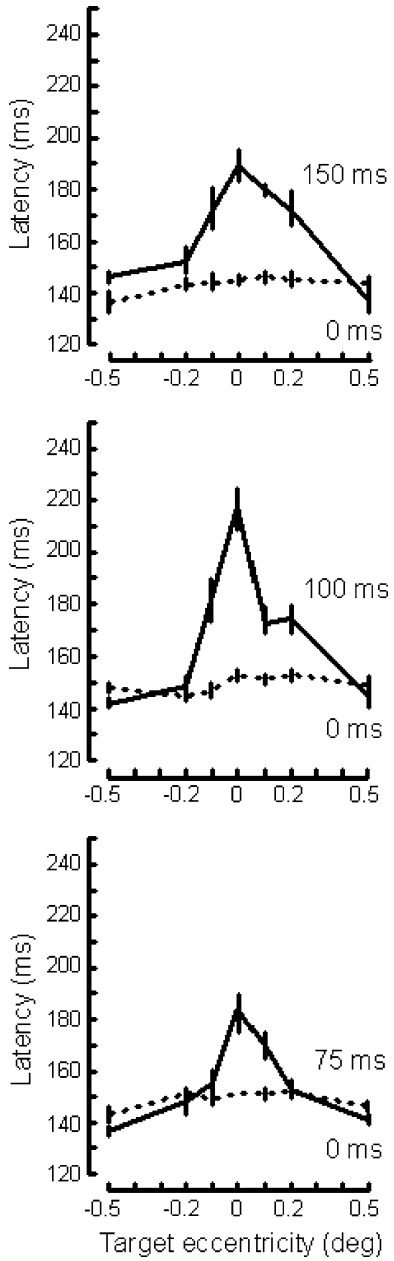

Fig. 6 Effect of stimulation on saccadic latency for different target positions at and outside of the receptive field of the stimulated neurons using three levels of train duration: 150 (*top*), 100 (*middle*), and 75 ms (*bottom*). Within each panel (*top*, *middle*, or *bottom*), the latency of visually guided saccades to the target is plotted as a function of target eccentricity with respect to the receptive field of the stimulated neurons. A zero eccentricity along the *x*-axis indicates that the target and the receptive-field centre of the stimulated neurons were in register (see Fig. 1c). *Negative values* along the *x*-axis indicate target positions situated between the fixation position and receptive-field centre of the stimulated neurons. *Positive values* indicate target positions eccentric to the receptive-field centre of the stimulated neurons. The *solid curve* represents data from stimulation trials and the *dotted curve* represents data from non-stimulation trials. Each value is based on 20 trials. Standard errors are shown. The depth of stimulation was 1.75 mm below the cortical surface. The stimulation included 100 μA anode-first pulses (at 0.2 ms duration) delivered at 200 Hz. The receptive field of the stimulated units was at 233° of meridian and at 3.4° of eccentricity. The target used was brighter than background at 33% contrast and was 0.2° in diameter. Data from monkey C. For other details see Materials and methods and Fig. 1b, c

for the largest target (0.2°) and greatest for the smallest target (0.1°). For the four sites tested (Fig. 9a–d), the size of the delay field was least for the largest target, intermediate for the mid-sized target, and greatest for the smallest target, respectively: a: 0.20° versus 0.29° versus 0.41°; b: 0.21° versus 0.31° versus 0.46°; c: 0.29° versus 0.29° versus 0.51°; d: 0.29° versus 0.29° versus 0.33°. The data in Fig. 9a were derived from Fig. 8.

Discussion

In a previous study, we found that delay fields are roughly circular and increase in size with increases in the eccentricity of the receptive field centres of the directly stimulated V1 cells (Tehovnik et al. 2005). In the present study we show that increases in current, pulse frequency, or train duration results in an increase in the size of a delay field. Also decreases in the size of the visual target tended to increase the size of a delay field. We now discuss the significance of these results.

Electrical parameters of stimulation affecting delay fields and phosphenes

When electrical stimulation is delivered to V1 of humans the subjects report the appearance of a punctate circular spot of light situated in that part of the visual field that corresponds to the site of stimulation in the V1 map (Brindley and Lewin 1968; Schmidt et al. 1996). There is a general consensus that increases in the current, pulse frequency, or train duration of stimulation delivered to V1 increase the brightness of a phosphene (Dobelle and Mladejovsky 1974; Rushton and Brindley 1978; Evans et al. 1979; Schmidt et al. 1996). However, less consistent and complete data are available on the effects of these parameters on phosphene size. Using surface stimulation through macroelectrodes, Rushton and Brindley (1977, 1978) found that increases in current resulted in an increase in phosphene size; whereas using depth stimulation through microelectrodes, Schmidt et al. (1996) reported that increases in current often led to a decrease in phosphene size. Schmidt et al. (1996) did find, however, that increases in train duration produced an increase in phosphene size.

In the present study, increases in current, pulse frequency, or train duration always resulted in an increase

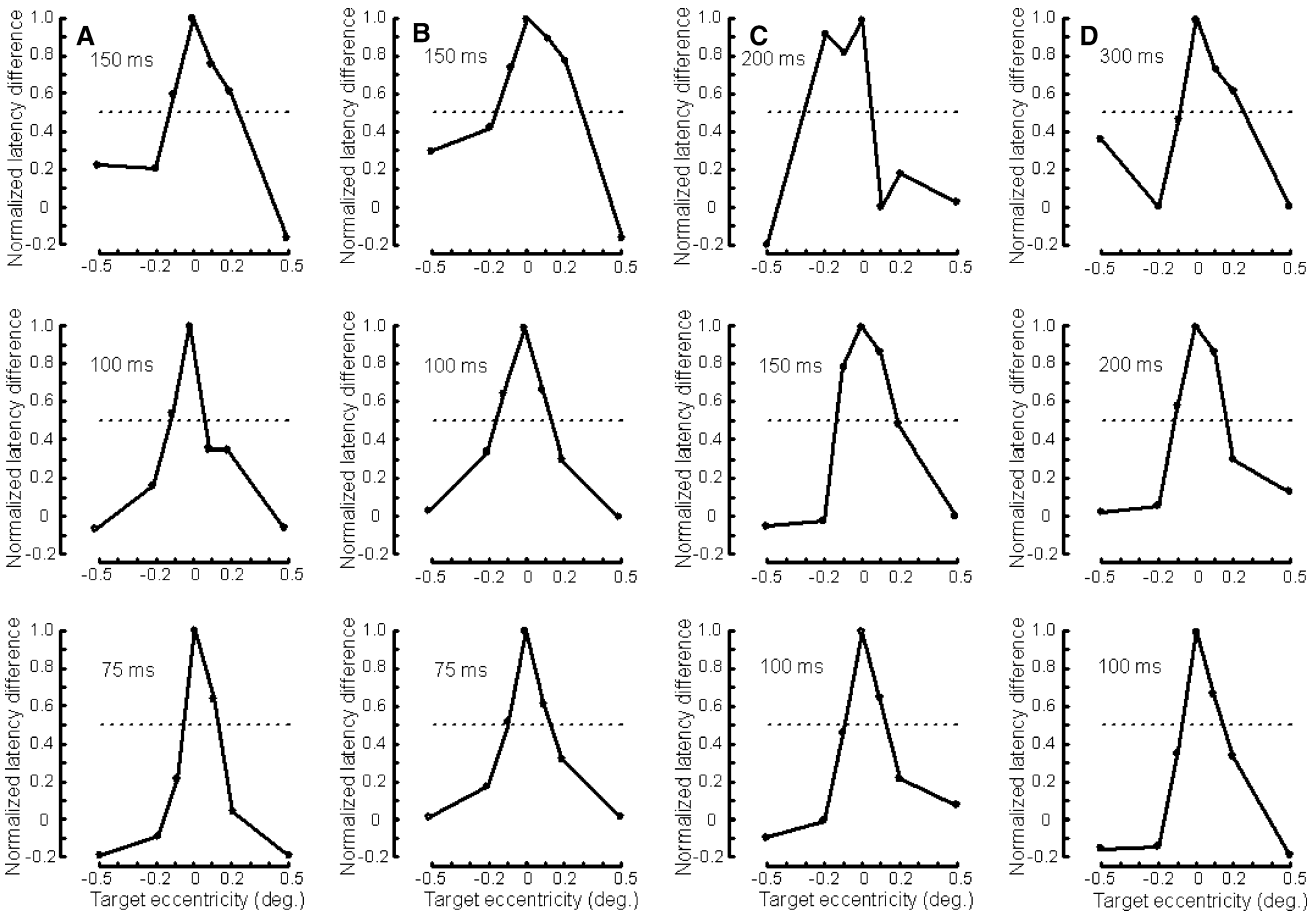

Fig. 7 Normalized latency difference is plotted as a function of target eccentricity for four stimulation sites (**a**–**d**) using three levels of train duration, 150, 200, or 300 ms (*top*), 100, 150, or 200 ms (*middle*), and 75 or 100 ms (*bottom*). Data in **a** are derived from Fig. 6. Each curve in a panel was determined by subtracting the latency of saccades during non-stimulation trials from the latency of saccades during stimulation trials to yield a latency difference. The latency-difference data were then normalized such that the values spanned the range from zero to one. Normalized latency-difference values less than zero indicate that the saccadic latencies during stimulation were less than those on non-stimulation trials. The size of a delay field was measured by noting the negative and positive target-eccentricity values at 50% of maximal latency difference represented by the *dotted horizontal line*. See Fig. 3 for other details. The size of the delay field was greatest for the longest train duration, intermediate for the intermediate train duration, and least for the shortest train duration: (**a** 0.37° vs. 0.19° vs. 0.17°; **b** 0.44° vs. 0.29° vs. 0.21°; **c** 0.36° vs. 0.32° vs. 0.21°; **d** 0.34° vs. 0.27° vs. 0.23°). For all sites, the stimulation was made up of 100 μA anode-first pulses (at 0.2 ms duration) delivered at 200 Hz. The target used was brighter than background at 33% contrast and was 0.2° in diameter. All data from monkey C. For other details see Materials and methods and Fig. 1b, c

in the size of a delay field. More studies need to be done on human (as well as monkey) V1 to establish whether manipulating current, pulse frequency, and train duration have incremental (or decremental) effects on phosphene size when stimulation is delivered through microelectrodes inserted into cortex.

Inhibitory circuits mediating the delay effect

The delivery of a single stimulation pulse to V1 renders neurons in V1 and in the lateral geniculate nucleus unresponsive to a visual stimulus for many tens of milliseconds after eliciting an initial burst (Schiller and Malpeli 1977; Chung and Ferster 1998). The hyperpolarizing effect of electrical stimulation is believed to be mediated by γ-aminobutyric acid (GABA)ergic interneurons (Krnjević et al. 1966a, b, c; Krnjević and Schwartz 1967; Douglas and Martin 1991; Butovas and Schwarz 2003). V1 stimulation might be delaying visually guided saccades by indirect activation of inhibitory interneurons following the direct excitation of pyramidal neurons. This idea is supported by three observations: (1) injection of the GABAergic agonist muscimol into V1 interrupts visually guided saccadic eye movements (Newsome et al. 1985; Schiller and Tehovnik 2003); (2) the excitability of the directly stimulated elements mediating the delay of saccades is similar to that of pyramidal fibres (Tehovnik et al. 2004); and (3)

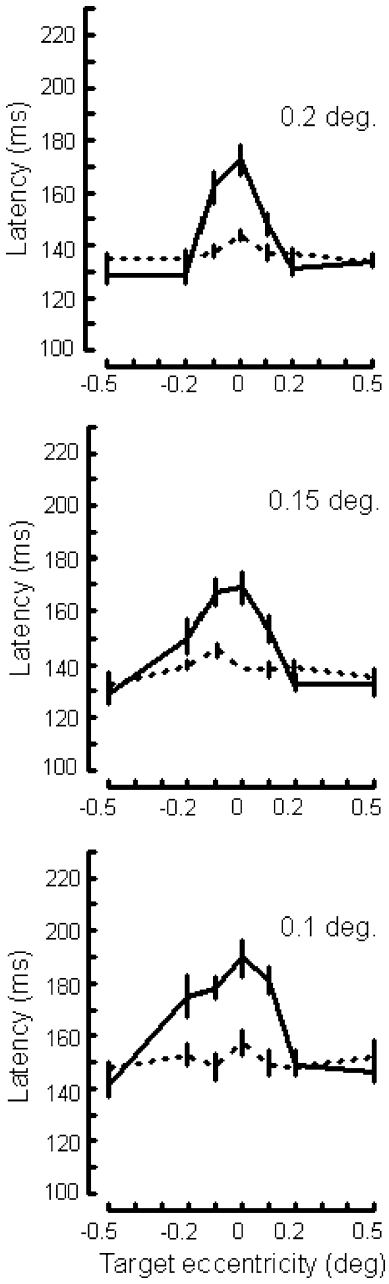

Fig. 8 Effect of stimulation on saccadic latency for different target positions at and outside of the receptive field of the stimulated neurons using three levels of target size, 0.2° (*top*), 0.15° (*middle*), and 0.1° (*bottom*). Within each panel (*top*, *middle*, or *bottom*), the latency of visually guided saccades to the target is plotted as a function of target eccentricity with respect to the receptive field of the stimulated neurons. A zero eccentricity along the *x*-axis indicates that the target and the receptive-field centre of the stimulated neurons were in register (see Fig. 1c). *Negative values* along the *x*-axis indicate target positions situated between the fixation position and receptive-field centre of the stimulated neurons. *Positive values* indicate target positions eccentric to the receptive-field centre of the stimulated neurons. The *solid curve* represents data from stimulation trials and the *dotted curve* represents data from non-stimulation trials. Each value is based on 20 trials. Standard errors are shown. The depth of stimulation was 2 mm below the cortical surface. The stimulation included 100 μA anode-first pulses (at 0.2 ms duration) delivered at 200 Hz using a 100 ms train. The receptive field of the stimulated units was at 223° of meridian and at 3.6° of eccentricity. The target used was brighter than background at 33% contrast. Data from monkey C. For other details see Materials and methods and Fig. 1b, c

single-pulse stimulation of V1 evokes a burst of activity followed by a 100–200 ms period of hyperpolarization within V1 pyramidal fibres (Douglas and Martin 1991). V1 pyramidal fibres are known to innervate GABAergic interneurons (Jones et al. 1994), which are believed to hyperpolarize pyramidal fibres locally via a feedback connection (Douglas and Martin 1991).

The excitability of neurons mediating the delay of saccades coincides with that of neurons mediating phosphenes in human V1 (Brindley and Lewin 1968; Dobelle and Mladejovsky 1974; Rushton and Brindley 1978; Tehovnik et al. 2004). Therefore, the delay of saccades might be a by-product of phosphene induction such that the pyramidal fibres are activated by the stimulation first (thereby producing a phosphene) followed by activation of GABAergic interneurons that inhibit the pyramidal elements thereby causing a saccadic delay.

Another explanation for the saccadic delay might be that once an animal detects the delivery of electrical stimulation to V1 (Bartlett et al. 2005; DeYoe et al. 2005), it suppresses the generation of a saccade to the receptive field location of the stimulated neurons. Suppression of saccades prevents against a break in fixation, which would lead to a forfeiture of the reward. The act of suppressing saccades may yield a delay in a subsequent saccade.

Effect of target size on delay fields

By changing the target size from 0.1 to 0.2° of visual angle, it was found that the size of the delay field decreased. So, why does the delay field get smaller when larger targets are used? If the delay effect is a consequence of the stimulation activating inhibitory circuits (as argued above), then the larger target may be more effective at overcoming the inhibition by driving the cells about the electrode tip at a higher rate. Indeed, when the visual target is greater than 0.3° of visual angle, the delay effect is absent (Tehovnik et al. 2004).

Are the inhibitory effects of V1 microstimulation related to vision?

It has been argued that the inhibitory inputs to simple and complex cells are only engaged once these cells have been activated and that the inhibition prevents

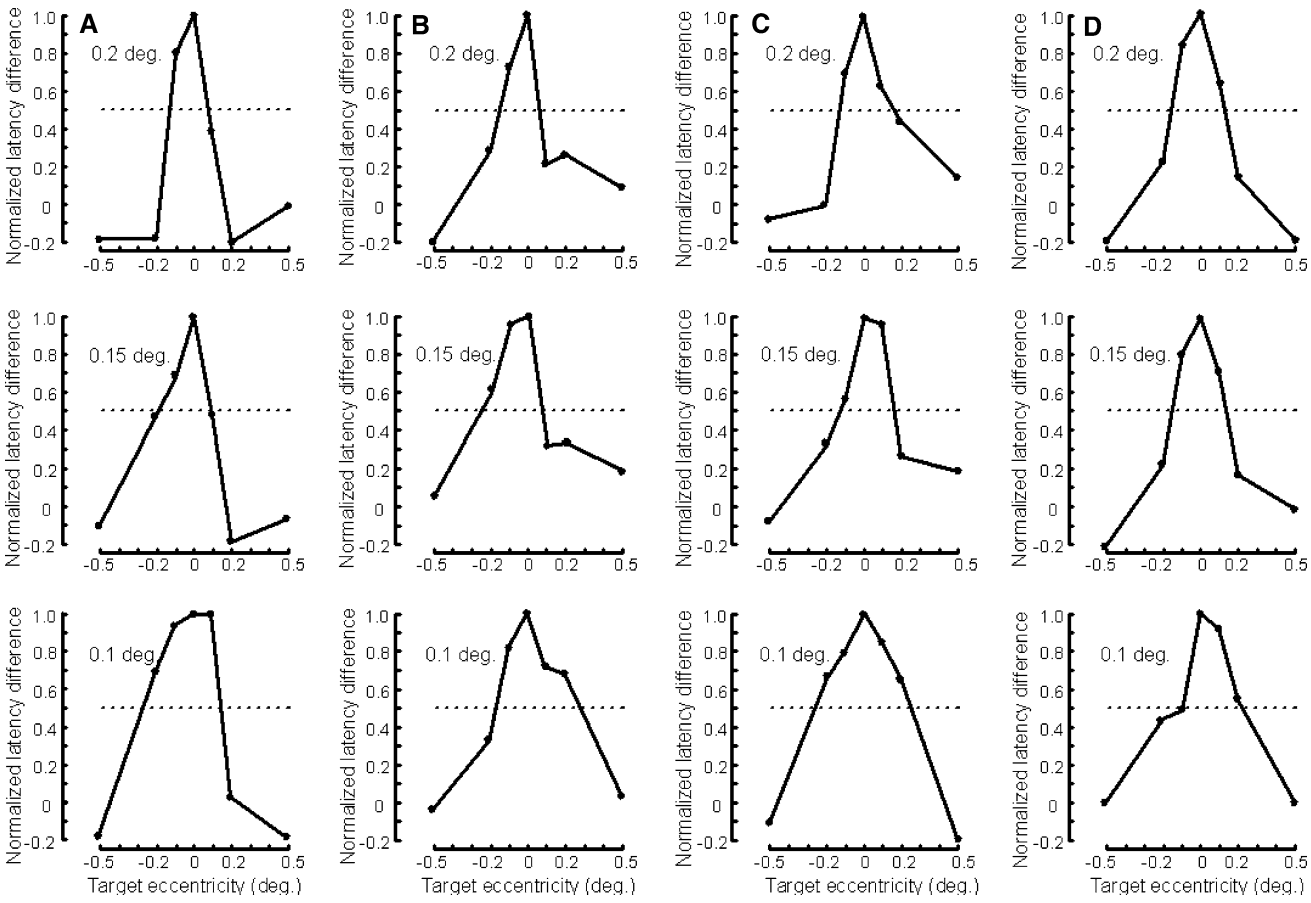

Fig. 9 Normalized latency difference is plotted as a function of target eccentricity for four stimulation sites (**a**–**d**) using three levels of target size, 0.2° (*top*), 0.15° (*middle*), and 0.1° (*bottom*). Data in **a** are derived from Fig. 8. Each curve in a panel was determined by subtracting the latency of saccades during non-stimulation trials from the latency of saccades during stimulation trials to yield a latency difference. The latency-difference data were then normalized such that the values spanned the range from zero to one. Normalized latency-difference values less than zero indicate that the saccadic latencies during stimulation were less than those on non-stimulation trials. The size of a delay field was measured by noting the negative and positive target-eccentricity values at 50% of maximal latency difference represented by the *dotted horizontal line*. See Fig. 3 for other details. The size of the delay field was least for the largest target, intermediate for the intermediate target, and most for the smallest target: (**a** 0.20° vs. 0.29° vs. 0.41°; **b** 0.21° vs. 0.31° vs. 0.46°; **c** 0.29° vs. 0.29° vs. 0.51°; **d** 0.29° vs. 0.29° vs. 0.33°). For all sites, the stimulation consisted of 100 μA anode-first pulses (at 0.2 ms duration) delivered at 200 Hz using a 100 ms train. The target used was brighter than background at 33% contrast. All data from monkey C. For other details see Materials and methods and Fig. 1b, c

against their unabated excitation (Douglas and Martin 1991) as seems to occur when the GABAergic antagonists bicuculline is applied to them (Sillito 1975). This supposition concurs with the finding that the excitatory and inhibitory inputs to simple and complex cells exhibit the same orientation tuning (Ferster 1986; Douglas et al. 1991) and that these inputs to simple cells have the same direction tuning despite differences in relative timing (Priebe and Ferster 2005). What this means is that once V1 cells discharge to particular features of visual stimulation, inhibitory neurons hyperpolarize the cells to prevent against their re-excitation. Hence, the electrical activation of these cells might delay saccadic eye movements generated to the visual stimuli coded by these cells. For such effects to be realized, however, currents below 10 μA might be necessary so that select groups of V1 cells coding for common visual attributes might be activated (Tehovnik and Slocum 2005). For example, Schmidt et al. (1996) found that in order to evoke coloured phosphenes, currents below 10 μA were required. Such currents can directly activate neurons within a V1 column coding for colour (Michael 1981; Tehovnik et al. 2006) thereby increasing the changes of eliciting a coloured phosphene.

Electrically evoked versus TMS-elicited interference

Much like electrical stimulation, transcranial magnetic stimulation (TMS) affects the brain by electrical excitation of neural circuits (Cowey 2005). The magnetic

field generated by TMS produces an electrical current in the brain. It is therefore not surprising that the delivery of electrical stimulation to V1 in monkeys produces effects that are similar to those observed following the delivery of TMS to V1 of humans. The major difference between the two methods is that electrical stimulation can be delivered through indwelling microelectrodes to better restrict the field of activation, whereas TMS is delivered through the skull with a less restricted field of activation but is more amenable to human use.

TMS over the occipital cortex of humans interferes with visual perception or induces phosphenes (Barker et al. 1985; Amassian et al. 1989; Áfra et al. 1998; Aurora et al. 1998; Kastner et al. 1998; Kammer et al. 2001). The field strength for inducing interference of a flashed punctate visual target is higher than the strength for eliciting phosphenes (Kastner et al. 1998; Kammer 1999; Kammer et al. 2005b). The phosphenes are described as whitish flashes but sometimes they are coloured (Áfra et al. 1998; Aurora et al. 1998; Kastner et al. 1998; Kammer 1999, 2001; Gothe et al. 2002). Both the interference and the phosphenes are confined to the same part of the visual field and the part of the visual field affected by TMS roughly corresponds to the site of stimulation over V1 (Kastner et al. 1998; Kammer 1999). It is believed that the locus of activation during TMS over the occipital cortex is mainly area V1 since phosphenes could not be evoked in a patient with lesions of V1 (Cowey and Walsh 2000).

The best interference occurs in humans when TMS is applied some 100 ms after target onset (Kastner et al. 1998; Kamitani and Shimojo 1999; Kammer 1999; Corthout et al. 2001; Kammer et al. 2005a) even though interference can occur when TMS is delivered at target onset (Corthout et al. 1999; Kammer et al. 2005a). The best saccadic delay (in monkeys) occurs when stimulation is delivered before and up until the onset of the visual target (current study; Tehovnik et al. 2004). Determining whether the interference observed in humans and monkeys is comparable will require that TMS experiments be performed on monkeys.

Conclusion

Our study shows that the size of a delay field increases with increases in current, pulse frequency, or train duration and decreases with increases in target size. We suggest that the delay effect is due to activation of inhibitory circuits recruited after the initial excitation of area V1 by electrical microstimulation although other explanations cannot be ruled out.

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